AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14 / April / 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

- Total Marks **60** marks
- This exam consists of **Three parts**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1- **Which of the following "V's" of Big Data refers to the speed at which data is generated and processed?**
   a- Volume
   b- Velocity
   c- Veracity
   d- Variety

2- **The role responsible for designing and maintaining the overall structure of databases in an organisation is:**
   a- Data Analyst
   b- Network Engineer
   c- Database Administrator
   d- Data Architect

3- **Which activity involves discovering patterns and insights from large datasets using statistical and machine learning techniques?**
   a- Data Collection
   b- Data Cleansing
   c- Data Storage
   d- Data Mining

4- **In the following data set: { 5, 8, 5, 12, 8, 5, 20 }, the mode is:**
   a- 8
   b- 12
   c- 20
   d- 5

5- **_____ is the process of combining data from different sources into a unified view.**
   a- Harmonization
   b- Integration
   c- Validation
   d- Normalization

6- **The technique of combining multiple datasets to re-identify individuals who were supposedly anonymous is called:**
   a- Data Encryption
   b- Linkage Attack
   c- Data Compression
   d- Indexing

7- **In a relational database, a Foreign Key:**
   a- Can contain NULL values
   b- Must be unique for every row
   c- Must be a numeric value
   d- Cannot reference another table

8- **A table is in _____ if it is in 2NF and all non-key attributes are non-transitively dependent on the primary key.**
   a- First Normal Form (1NF)
   b- Second Normal Form (2NF)
   c- Third Normal Form (3NF)
   d- Boyce-Codd Normal Form

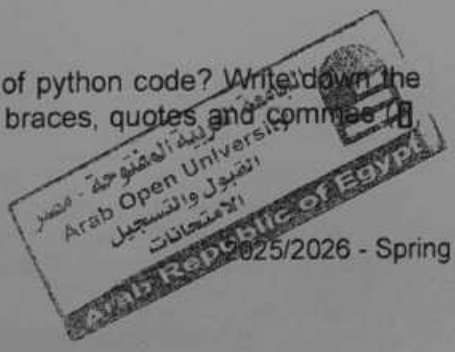
9- **A Materialized View differs from a standard view because it:**
   a- Is computed every time it is queried
   b- Is physically stored on disk with pre-computed results
   c- Cannot be updated
   d- Does not require a base table

10- **The SQL command used to permanently remove all rows from a table while keeping its structure is:**
   a- DROP
   b- REMOVE
   c- DELETE
   d- TRUNCATE

Part 2: Programming and Database Questions [40 marks]

Question One: [10 marks]

What will be the output of the following snippets of python code? Write down the exact output, including all brackets, parentheses, braces, quotes and commas (0, (), {}, '', "", ,) . (2 Marks for each question)



1.

```
def classify(n):
    if n % 3 == 0:
        return n * 3
    else:
        return n + 100

num = 9
output = classify(num)
print("Result for", num, "is:", output)
```

2.

```
cities = ["Muscat", "Dubai", "Riyadh"]
cities.insert(1, "Doha")
for i in range(len(cities)):
    if i == 2:
        print("Third city:", cities[i])
```

3.

```
record = {"EmpID": 202, "Dept": "HR", "Leave": 12}
record["Leave"] = record["Leave"] - 3
print(record["EmpID"], "has", record["Leave"], "leave days")
```

4.

```
values = [3, 5, 7]
total = 0
for v in values:
    total += v * 2
print("Total is:", total)
```

5.

```
class Phone:
    def __init__(self, brand, price):
        self.brand = brand
        self.price = price

p1 = Phone("Samsung", 300)
p1.price += 50
print(p1.brand + "Pro")
print(p1.price - 100)
```

Question Two: [20 marks]

A university needs a database to track Student Enrollments in various Courses.

- **Task 1:** Normalize the data into 1NF, 2NF, and 3NF. Show the relations for each form. **[10 Marks]**
- **Task 2:** List Primary Keys and Foreign Keys for the 3NF tables in the format: **Table Name | Primary Key | Foreign Key**. **[10 Marks]**

Stu ID	Stu_Name	Course_ID	Instructor	Module_Details
S01	Ali Hassan	CS101	Dr. Noor	{ Module_Code: M1, Title: Databases, Credits: 3 } { Module_Code: M2, Title: Python, Credits: 2 }
S02	Maha Salem	CS102	Dr. Fans	{ Module_Code: M3, Title: Networking, Credits: 3 }

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Question Three: [10 marks]

Consider the following schema:

- **STUDENT** (**stu_id**, stu_name, major, gpa)
- **COURSE** (**course_id**, course_name, credits)
- **ENROLLMENT** (**stu_id, course_id**, grade, enroll_date)

Write SQL queries for:

1. List the name and GPA of all students with a major of 'Computer Science' and GPA > 3.0. (3 Marks)
2. List the names and credits of all courses that have at least one student enrolled in them. (3.5 Marks)
3. Update the GPA of all students in the 'Engineering' major by increasing it by 0.2. (1.5 Marks)
4. Calculate the total number of enrollments for each course (group by course_id). (2 Marks)

Part 3: Data Format Questions [10 marks]

Question One [5 Marks]

Convert this XML to JSON.

XML

```
<university>
  <staff>
    <id>ST201</id>
    <n>Dr. Layla Al-Habsi</n>
    <role>Lecturer</role>
  </staff>
  <courses>
    <course>Data Management</course>
    <course>Web Development</course>
  </courses>
</university>
```

Question Two [5 Marks]

Convert this JSON to XML.

JSON

```
{
  "invoiceID": "INV-88",
  "client": "Sara",
  "products": ["Keyboard", "Monitor"],
  "shipping": {
    "carrier": "DHL",
    "fee": 8
  }
}
```

End of the questions

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